Is a class-group 2-rank of 6 unusual for a prime-order curve's CM discriminant? (P-224)

Date 2026-10-22. Frozen protocol: `PROTOCOL.md` (committed before the run; SHA-256 `7b857376a897560b30ed964b0ddc7e064798227969b4db70475a5b310268090e`). Code: `src/cryptanalysis/two_rank_null.rs`, `examples/p224_two_rank_null.rs`. Raw rows: `run/rows.jsonl` (6822 rows); everything below is recomputed from them by `analyze`. Seed 20261022, trial bound 10^6, Pollard-Brent budget 2^{21} iterations, 4 threads, 54.6 min wall, 127 CPU-min (estimate: 60 min wall). This is a structural statistic, not an ECDLP cost measurement: no rho ratio, no scoreboard row, no improvement of the index-calculus baseline.

Answer (frozen decision rule, measured)

P-224's $\omega(D_K) = 7$ (2-rank 6) is within the control distribution. Against random 224-bit primes (R224u, N = 1000) 12.3% of controls have $\omega \geq 7$ (Clopper-Pearson 95%: 10.3% to 14.5%); the mid-rank percentile of 7 is 91.5% (control median 4, 2.5% to 97.5%: 2 to 8). Against the fixed P-224 prime (F224u) 14.5% (12.4% to 16.8%) of controls reach 7, mid-rank 90.0%. Two-sided add-one p (minimum over censoring versions) 0.136 and 0.162, against the corrected threshold $\frac{\alpha}{m} = 0.005$. Effect size: P-224 is 3 above the control median of 4 (2 above the fixed-prime median of 5) (null mean 4.47). The part of ω made of primes $\leq 10^6$ (ω_{small}, never censored) is 5 for P-224 against a control median of 2 (95.5th mid-rank percentile in R224u, upper tail 6.9%, 5.4% to 8.7%); also within the control distribution at the corrected threshold ($p = 0.140$, and 0.162 for F224u).

P-256 is inconclusive under the frozen instrument (its 258-bit discriminant leaves a 215-bit unsplit cofactor: $\omega \in [4, 13]$), and its ω_{small} (2) is at the control median (50.2%). The published factorisation (5 primes) puts $\omega = 5$ at a point 48.9% of R256u controls reach or exceed on ω_{mid} (post hoc, below).

Class of each statement: ω values of P-224, brainpoolP224r1, brainpoolP256r1, secp224k1, secp256k1: exact computation (complete factorisations, product and primality checked by the code). Percentiles and verdicts: comparative, from this controlled experiment. The 2-rank identity: theorem (genus theory).

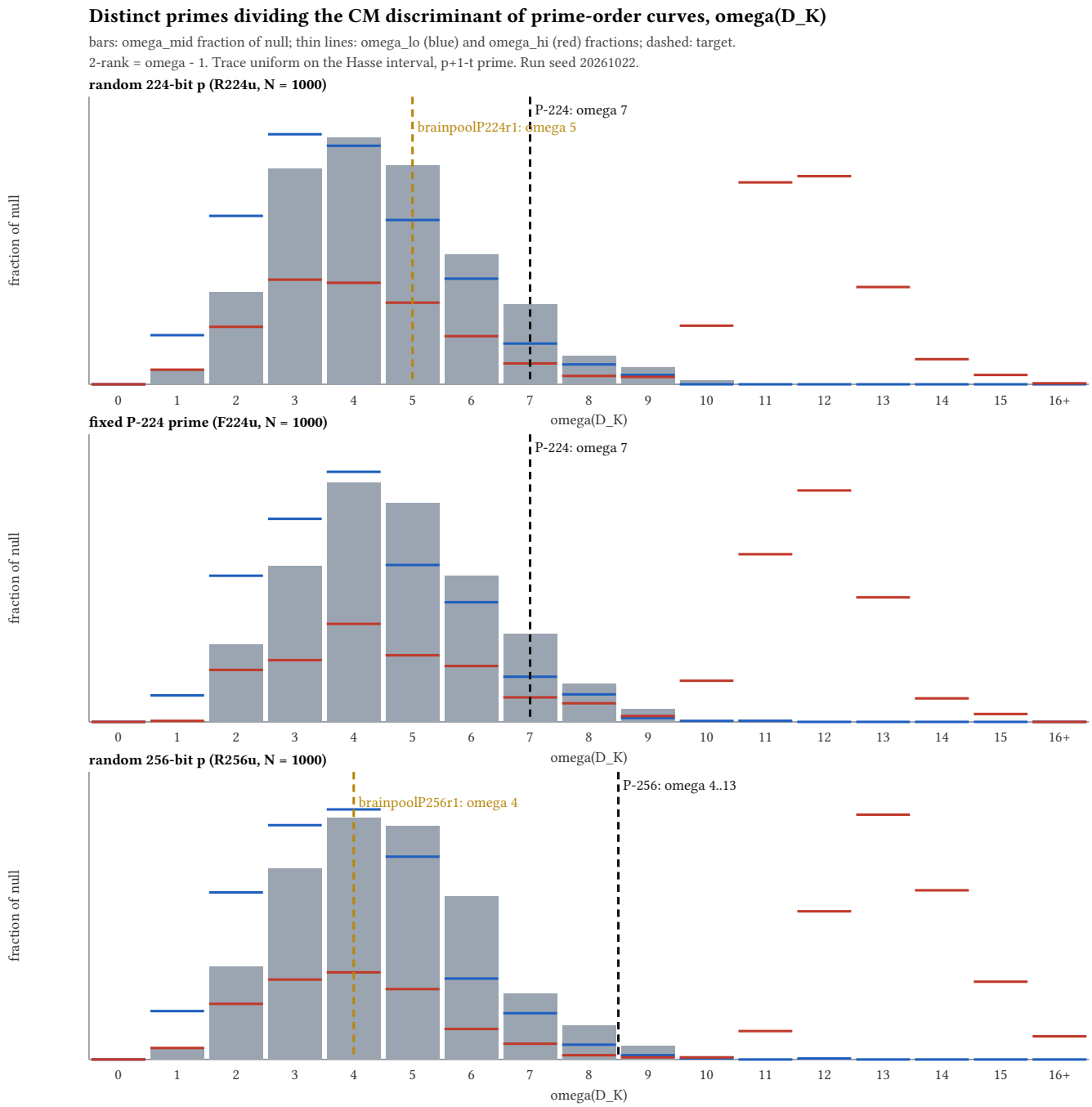


Figure 1: Null distribution of $\omega(D_K)$ for the three uniform-trace primary arms, N = 1000 each (run `20261022`). Grey bars: ω_{mid}; blue and red lines: the rigorous lower (ω_{lo}) and upper (ω_{hi}) versions. Dashed: targets (P-256 is at the midpoint of its bound 4 to 13). Source rows: `run/rows.jsonl`.

Null arms and censoring

Censoring is the fraction of samples whose discriminant keeps an unsplit composite cofactor after the shared budget. Those samples were kept and bounded, never dropped.

arm	N	censored (95% CP)	fully factored	ω_{mid} median [2.5%, 97.5%]	ω_{small} median [2.5%, 97.5%]
R224u	1000	577 = 57.7% (54.6 to 60.8)	423	4 [2, 8]	2 [0, 5]
R224s	1000	596 = 59.6% (56.5 to 62.7)	404	4 [2, 8]	2 [0, 5]
F224u	1000	611 = 61.1% (58.0 to 64.1)	389	5 [2, 8]	2 [0, 6]
F224s	1000	597 = 59.7% (56.6 to 62.8)	403	5 [2, 8]	2 [0, 6]
R256u	1000	658 = 65.8% (62.8 to 68.7)	342	4 [2, 8]	2 [0, 5]
R256s	1000	663 = 66.3% (63.3 to 69.2)	337	4 [2, 8]	2 [0, 5]
R384u	100	80 = 80.0% (70.8 to 87.3)	20	5 [2, 9]	2 [0, 6]
R521u	100	85 = 85.0% (76.5 to 91.4)	15	5 [2, 9]	2 [0, 6]

The rigorous upper bound ω_{hi} is loose (median 11 to 13 for 224- and 256-bit arms), so a verdict of "unusual" on a censored metric is only reachable for extreme targets; "within the control distribution" is decided by the favourable bound $p_{\min}$. No overlap flag was raised in any arm. Samples never resampled or dropped: 0.

Pre-registered family (uniform trace, $\frac{\alpha}{m} = 0.005$, $m = 10$)

target	null	metric	value	control median [2.5%, 97.5%]	mid-rank pct; upper tail (95% CP)	p min / mid / max	verdict
P-224	R224u	ω	7	4 [2, 8]	91.5%; 12.3% (10.3 to 14.5)	0.136 / 0.248 / 1.000	within the control distribution
P-224	R224u	ω_{small}	5	2 [0, 5]	95.5%; 6.9% (5.4 to 8.7)	0.140	within the control distribution
P-224	F224u	ω	7	5 [2, 8]	90.0%; 14.5% (12.4 to 16.8)	0.162 / 0.292 / 1.000	within the control distribution
P-224	F224u	ω_{small}	5	2 [0, 6]	94.5%; 8.0% (6.4 to 9.9)	0.162	within the control distribution
P-256	R256u	ω	4 to 13	4 [2, 8]	62.2%; 48.9% (45.8 to 52.0)	0.002 / 0.979 / 1.000	inconclusive: censoring
P-256	R256u	ω_{small}	2	2 [0, 5]	50.2%; 63.5% (60.4 to 66.5)	1.000	within the control distribution
brainpoolP224r1	R224u	ω	5	4 [2, 8]	64.8%; 45.6% (42.5 to 48.7)	0.651 / 0.913 / 1.000	within the control distribution
brainpoolP224r1	R224u	ω_{small}	3	2 [0, 5]	73.2%; 36.2% (33.2 to 39.3)	0.725	within the control distribution
brainpoolP256r1	R256u	ω	4	4 [2, 8]	39.6%; 71.9% (69.0 to 74.7)	0.448 / 1.000 / 1.000	within the control distribution
brainpoolP256r1	R256u	ω_{small}	1	2 [0, 5]	23.5%; 89.5% (87.4 to 91.3)	0.731	within the control distribution

Sensitivity (semicircle-weighted trace, not in m): P-224 mid-rank 92.5% (R224s) and 88.8% (F224s) on ω, 96.3% and 94.6% on ω_{small}, all within the control distribution (minimum p 0.120 to 0.200); P-256 inconclusive, brainpool within, the same as the uniform arms. The conclusion does not depend on the trace scheme. Full tables: `run/summary.md`.

Descriptive comparison points (outside the tested family; no verdict)

target	measured under the same instrument	null	note
P-384	$\omega \in [2, 19]$, $\omega_{\text{small}} = 0$, unsplit cofactor 362 bits	R384u, N = 100	99% of controls have $\omega_{\text{hi}} \geq 2$; no percentile resolvable (power check fails)
brainpoolP384r1	$\omega \in [3, 18]$, $\omega_{\text{small}} = 1$, unsplit 326 bits	R384u	as above
P-521	$\omega \in [2, 27]$, $\omega_{\text{small}} = 1$, unsplit 521 bits	R521u, N = 100	as above
secp224k1, secp256k1	$\omega = 1$ (exactly)	R224u, R256u	$j = 0$: $\Delta = -3v^2$, $D_K = -3$ by CM theory. Not a member of the generic null; 4.7% and 4.6% of controls have $\omega_{\text{lo}} \leq 1$ only because the control population has no $j = 0$ curves. No percentile is claimed.

Positive control (run first): passed

(a) Population: 300 fresh R224u samples with k planted small primes through the same code path, Mann-Whitney against the R224u null. $k = 2$: mean ω_{mid} shift +2.05, $z = 15.9$, $p = 8 \times 10^{-57}$. $k = 4$: shift +4.05, $z = 23.8$, $p = 4 \times 10^{-125}$. Both detected (rule: $p < 0.001$ and shift within 0.5 of k). (b) Target sweep: P-224′s discriminant times the first k primes ≥ 5 not dividing it. On ω_{small} the detector flags (corrected $p_{\max} < 0.005$) from $k = 3$ ($\omega_{\text{small}} = 8$); on the censored ω it flags from $k = 9$ ($\omega = 16$, $p_{\max} = 0.004$; $k = 3$ to 8 are "inconclusive: censoring" because $p_{\max}$ reaches 1.0 to 0.022 while $p_{\min} = 0.002$). Pre-registered rule (flagged at some $k \leq 10$ on both metrics): **met**. Reading: the instrument flags a planted excess of 9 primes on the censored metric (planted ω = 16) and of 3 on the exact one; an excess of 3 over the control median on the censored ω (P-224′s actual excess) would not be flagged at this N by the rigorous bound, which is what "within the control distribution" with $p_{\min} = 0.14$ rests on, not a claim of equivalence.

Observations outside the frozen rule (post hoc, labelled, not part of any verdict)

1. P-256 with its published factorisation ($\omega = 5$, factors in `research/p224_class_number_20261022/PROTOCOL.md`): 48.9% of R256u controls have $\omega_{\text{mid}} \geq 5$ (unequal instrument: published factors for the target, budgeted rho for the controls, so no verdict).
2. Conditioning on being fully factored by the instrument (as P-224 is): 33 of 423 R224u controls (7.8%) have $\omega \geq 7$. Complete factorability selects smoother, higher-ω numbers, so this is the like-for-like upper-tail fraction for a P-224-like input; P-224 sits at the 92nd percentile of that subpopulation, not beyond it.

Scope: what this does not show

- The null is not a point-counted sample. N is drawn uniformly (and, as sensitivity, semicircle-weighted) from the Hasse interval subject to primality, with random or fixed p; `random b` curves were not generated because point counting is unavailable. Class-number weighting of traces is not modelled.
- ω of 58 to 66% of controls is bounded, not known. Verdicts on ω rest on the pre-registered bound rule; the favourable bound ($p_{\min}$) carries every "within" verdict.
- P-256, P-384, P-521 and brainpoolP384r1 have censored discriminants under this instrument, so no percentile claim is made for them (P-256 inconclusive; the others descriptive).
- The study is about the count of distinct primes (the genus-theory 2-rank). It says nothing about the 4-rank, the size of the class group, or the hardness of any discrete-logarithm problem. "Within the control distribution" is not "random" and not "secure".
- No earlier claim is weakened. PR 2408′s exact value (2-rank 6 for P-224, 4 for P-256, from complete published factorisations) is unchanged; this report only adds that 6 is not extreme for the population.